

# Problem Statement and Proposed Solution

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## 1. The problem

Citizens of Madhya Pradesh already have ways to complain: the CM Helpline, a toll-free number, email, WhatsApp and the Bhopal Municipal Corporation (BMC) portal. What they lack is a way to **describe a problem in their own words and have it reach the right office without knowing how the government is organised**.

### Evidence from Madhya Pradesh (Bhopal)

| Finding                                                                                                                                                                                                | Source                                                                                                                                   | Confidence                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| BMC is the civic body for Bhopal; head office at Harshwardhan Complex, Mata Mandir, Bhopal 462001                                                                                                      | <a href="http://bhopal.nic.in">bhopal.nic.in</a>                                                                                         | High, official                                         |
| Civic complaints (water, road, sewerage, streetlight) can be lodged through the CM Helpline, a toll-free number, email, WhatsApp or the BMC portal, and a complaint needs a <b>zone or ward number</b> | <a href="http://complainthub.org">complainthub.org</a>                                                                                   | Medium, unofficial guide                               |
| The structure is department → body → zone → ward. Ward count is quoted as 85 or 86 and zone count as 19 or 21 by different unofficial sources                                                          | <code>docs/research/T05-research.md</code>                                                                                               | Unverified, conflicting                                |
| Who owns a "no water" complaint is ambiguous: BMC distribution versus PHED for bulk supply                                                                                                             | <code>docs/ARCHITECTURE_DEPARTMENTS.md</code>                                                                                            | Ambiguous                                              |
| MPOne's own Customer Solutions Hub is a voice/call-centre-first platform with an executive dashboard; multilingual support is listed only as a future item                                             | <a href="http://eservicesnest.com">eservicesnest.com</a> (competitor research, <code>docs/COMPETITOR_RESEARCH_ESERVICES_NEST.md</code> ) | Their own marketing claims, not independently verified |

### What this means for the citizen

1. **Knowledge barrier.** The citizen must know the department and the ward or zone before filing.
2. **Language barrier.** Forms and menus assume standard written language; many people speak Hindi, Hinglish or a dialect.
3. **Misrouting.** Ownership is genuinely ambiguous in places, so complaints land in the wrong queue.
4. **No feedback.** Without a complaint number and a simple status check, the citizen cannot tell whether anything happened.

### What this means for officers

Raw, unstructured complaints must be read, classified and forwarded by hand before any work starts.

## 2. Proposed solution

A voice-first AI chatbot **website** that turns a spoken or typed complaint into a routed, trackable ticket:  
speak or type → understand → ask only what is missing → confirm → ticket SMD-xxxx at the right office → check status

| Need                           | How Samadhan answers it                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| No need to know the department | The AI picks the department from the citizen's words; when unsure it asks "is this X or Y?" with questions we wrote                              |
| Own language                   | Hindi and Hinglish text and voice (Sarvam speech-to-text, Groq Whisper fallback) and spoken replies (Sarvam Bulbul)                              |
| No need to know the ward       | Ward name matched fuzzily to the ward office; GPS is captured for the officer map                                                                |
| Wrong routing                  | Low confidence or unknown place goes to the district office as <code>needs_review</code> ; officers can reassign, and every correction is logged |
| Feedback                       | Complaint number SMD-xxxx; status by typing or saying the number in the chat, or on a status page                                                |
| Officer workload               | Tickets arrive classified, with department, office, a summary and the original audio, plus a review queue and a map                              |

## 3. Scope of the prototype

- **Departments:** water supply (the original pilot) plus electricity, roads, sanitation and a Human Evaluation desk for unclear cases. **Office data for the last four is DEMO** and labelled so. Ward names are unverified.
- **Coverage:** built for all of Madhya Pradesh. The demo data covers Bhopal (5 demo wards per department plus one district fallback office each). Adding a district means adding data rows, not code.
- **Channels:** website (text, in-browser voice, GPS). WhatsApp, calls and a mobile app are out of scope for now.
- **Information questions** ("how do I get an income certificate?"): a fixed honest reply, a validated `.gov.in` link and a disclaimer. Samadhan does **not** answer government-information questions itself and does **not** read or verify any document.

## 4. What we deliberately did not do

- No invented answers: the AI cannot create a department, field or office.
- No document handling: documents and government records need the citizen's consent and official integration (for example DigiLocker), which is roadmap work.
- No claim of measured resolution-time improvement. That needs real usage over time.